MIL-D-23296A (OS)
14 June 1972
SUPERSEDING
MIL-D-23296 (Wep)
9 May 1962

MILITARY SPECIFICATION

DIETHYLENE GLYCOL

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for diethylene glycol. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED-STD-791	Lubricants, Liquid Fuels and Related Products, Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
*MIL-STD-290	Packaging, Packing, and Marking of Petroleum and Related Products
MIL-STD-1341	Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials

ASTM D 270	Method of Sampling Petroleum and Petroleum Products
ASTM D 891	Methods of Test for Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
ASTM D 1078	Method of Test for Distillation Range of Volatile Organic Liquids

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. The diethylene glycol shall be a product so prepared as to meet the requirements of this specification (see 4.5).

3.2 Specific gravity. The specific gravity of the diethylene glycol shall be between 1.1170 and 1.1200 at 20°/20° centigrade (C) when determined as specified in 4.6.1.

3.3 Distillation range. The diethylene glycol shall have an initial boiling point of 242° C minimum and dry point of 250° C maximum when determined as specified in 4.6.2.

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3.4 Acidity. The diethylene glycol shall have an acidity of 0.005 percent maximum by weight as acetic acid when determined as specified in 4.6.3.

3.5 Moisture. The diethylene glycol shall have a moisture of 0.2 percent maximum by weight when determined as specified in 4.6.4.

3.6 Ash. The diethylene glycol shall have an ash of 0.005 percent maximum by weight when determined as specified in 4.6.5.

3.7 Color. The diethylene glycol shall have a color of 15 units maximum (platinum-cobalt) when determined as specified in 4.6.6.

3.8 Water solubility. The diethylene glycol shall be miscible in all proportions when determined as specified in 4.6.7.

3.9 Workmanship. The diethylene glycol ingredients shall be processed to meet the requirements of this specification (see 4.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance inspection. Conformance of the diethylene glycol to the requirements of this specification shall be determined entirely by means of acceptance inspection. The acceptance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.2) for all of the requirements of this specification, and an examination of the sample of filled containers (4.3.2) for conformance to the packaging, packing, and marking requirements of section 5.

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4.3 Sampling. The material shall be subject at all times to inspection at the place of manufacture. Each case or large container shall be carefully identified by the manufacturer's batch (or control lot) number, until ultimate action is taken as to acceptance or rejection of material offered for delivery. Individual samples shall not be mixed, shall be placed in separate airtight and watertight containers, which shall be nearly filled, covered, and sealed to prevent atmospheric effects, and shall be labeled completely with information on the lot and batch number, date of sampling, contract number, and applicable specification. A batch is defined as the end product of all the raw materials mixed or blended in a single operation.

4.3.1 Size of lot. For the purpose of sampling, a lot of diethylene glycol shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of not more than 2,000 gallons of diethylene glycol offered for delivery at one time.

*4.3.2 Sampling for the inspection of filled containers. A random sample of filled containers shall be taken from each lot in accordance with MIL-STD-105, at inspection level II, and acceptable quality level (AQL) of 2.5 percent defective.

4.3.3 Sampling for tests. Samples for tests shall be taken in accordance with ASTM method D 270.

4.4 Inspection. Inspection shall be performed in accordance with method 9601 of FED-STD-791.

4.4.1 Examination of filled containers. Samples taken in accordance with 4.3.1 shall be examined for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Any container having one or more defects or under the required fill shall be rejected. The lot represented by a sample shall be rejected if the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105.

4.5 Classification of tests. All tests are quality conformance tests. Conformance of the diethylene glycol to the requirements for materials (3.1) and workmanship (3.9) shall be determined by appropriate examination and testing in accordance with section 3.

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4.6 Test methods.

*4.6.1 Specific gravity. The specific gravity shall be determined at 20°/20° C ± 0.02° C in accordance with ASTM method D 891, method C.

4.6.2 Distillation range. The distillation test shall be conducted in accordance with ASTM method D 1078 using a gas burner as the source of heat. It is necessary to use the proper ASTM thermometer or an equivalent thermometer of suitable range, calibrated for 100-millimeter immersion, graduated in 0.2° C units, and capable of being read to 0.1° C, easily.

4.6.3 Acidity. Acidity shall be determined in accordance with the following procedure: Accurately weigh a 50-gram sample of diethylene glycol into a 250-milliliter (ml) Erlenmeyer flask, add 100 ml methyl alcohol, mix, and titrate with 0.1 N NaOH using phenolphthalein indicator. Titrate a blank containing 100 ml of methanol. Subtract the blank titration from the amount of NaOH used for the sample titration and use this difference in the calculation.

$$\text{Percent acetic acid} = \frac{(\text{ml NaOH})(N)(6.005)}{W}$$

where

N = normality of NaOH
W = weight of sample.

4.6.4 Moisture. The moisture content of the diethylene glycol shall be determined in accordance with the procedure described in FED-STD-791, method 3255. If desired, a stabilized single solution may be substituted for the 2-solution Karl Fischer reagent.

*4.6.5 Ash. Ash shall be determined in accordance with the following procedure: Heat a platinum, porcelain, or silica crucible to redness, cool, and weigh. Reheat to constant weight so that repeated weighings are within ±0.1 milligram (mg). Place approximately 20 grams of the diethylene glycol in the crucible and weigh to the nearest 0.1 gram. Heat gently with a Bunsen burner until the material can be ignited at the surface. Remove the burner and allow the material to burn completely. After all the free carbon on the sides of the crucible has burned completely, heat with a strong flame or in a muffle furnace at 700° - 750° C (1292° - 1382° F) until all carbonaceous material has disappeared. Cool

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and weigh the crucible. Repeat the heating, cool, and weigh until constant weight within ±0.1 mg is obtained.

$$\text{Percent ash} = \frac{\text{Weight of residue} \times 100}{\text{Weight of sample}} .$$

4.6.6 Color. The color shall be determined by comparison with Hazen or APHA platinum-cobalt standards. The color standards are dilute HCl solutions of platinum and cobalt in fixed ratios. They are used for measuring the color of essentially water-white liquids and are applicable only to liquids in which the color-producing bodies present have light absorption characteristics nearly identical with those of the platinum-cobalt standards. The same proportions of platinum and cobalt are used to prepare Hazen and APHA color standards. These are the same both numerically and in concentration.

4.6.6.1 Preparation of color standards. Dissolve 1.245 grams of potassium chloroplatinate (K_2PtCl_6) and 1.000 gram of crystallized cobalt chloride ($CoCl_2 \cdot 6H_2O$) in distilled water containing 100 ml of 37 percent HCl and make up to 1000 ml in a volumetric flask with distilled water. This solution will have a Hazen or APHA color of 500. (This solution may also be purchased from Fisher Scientific Company, New York City, under the designation Platinum-Cobalt Color Standard Solution Number 500 or APHA Catalog No. SO-P-120 in 4-, 16-, or 32-ounce quantities.) Prepare the lighter color standards by diluting the 500 standard with distilled water. This is best done in two steps for colors below 50 in order to avoid measurement of extremely small volumes of the color standard. See table I for dilution ratios. Store the color standards in glass-stoppered bottles, preferably Pyrex. These standards are permanent for periods of 1 to 2 years if stored in tightly stoppered bottles.

4.6.6.2 Procedure. Compare the color of the sample visually with an equal depth of the color standards by looking down through the liquid column. It is recommended that the comparison be made in Nessler tubes; however, it can be performed in any uniformly shaped, clean, glass bottles using similar bottles for the standards. Report as the color the number of the standard that most nearly matches the sample. In the event that the color lies midway between two standards, report the darker of the two.

4.6.7 Water solubility. Water solubility shall be determined as follows: Measure 25 ml of the diethylene glycol at 77° ± 3° F (25° ± 2° C) into a 100-ml, glass-stoppered graduate, and add 25 ml of distilled water

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at 77° ± 3° F (25° ± 2° C) in 5-ml portions. Shake the graduate well after each addition. Observe the results after each shaking. In another graduate add 25 ml of the diethylene glycol to 25 ml of distilled water and treat in the same manner. The sample is completely miscible if there is no cloudiness or turbidity at any time.

Table I

HAZEN OR APHA COLOR STANDARDS

Color standards	500 STD (ml)	Distilled water (ml)
150	30	70
100	20	80
50	10	90
Color standards	**50 STD (ml)**	**Distilled water (ml)**
50	100	0
40	80	20
30	60	40
20	40	60
15	30	70
10	20	80
5	10	90
0	0	100

*5. PREPARATION FOR DELIVERY

*5.1 Packaging, packing, and marking. Unless otherwise specified in the contract or purchase order (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290. Each package shall be labeled with the hazardous identification symbol in accordance with MIL-STD-1341.

*5.2 Safety data sheet. When required by the contract or purchase order, the supplier shall prepare a safety data sheet in accordance with MIL-STD-1341. The safety data sheet shall be submitted with each shipment as specified in MIL-STD-1341.

6. NOTES

*6.1 Intended use. The diethylene glycol covered by this specification may safely be used in ordnance applications as a desensitizer for a liquid nitrate ester and to lessen the hazard of transporting this material.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Quantity of diethylene glycol required in terms of U. S. gallons

(c) Type and size of container (see 5.1)

(d) Selection of applicable levels of packaging and packing and other options, choices, and alternatives in MIL-STD-290.

*6.2.1 Deliverable data items. Items of deliverable data required by this specification are cited in the following paragraphs herein:

Paragraph	Data requirement	Applicable DID
5.2	Safety data sheet	-

DID's (Data Item Descriptions/DD Form 1664) for the above data requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into the contract.

*6.3 Batch. A batch is defined as that quantity of diethylene glycol that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

6.4 Changes from previous issue. The outside margins of the documents have been marked with asterisks to indicate where changes, deletions, or additions from the previous issues have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notation and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-D-23296A (OS), Diethylene Glycol

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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